

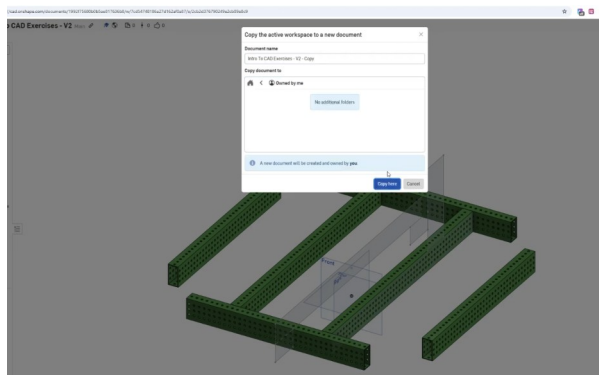
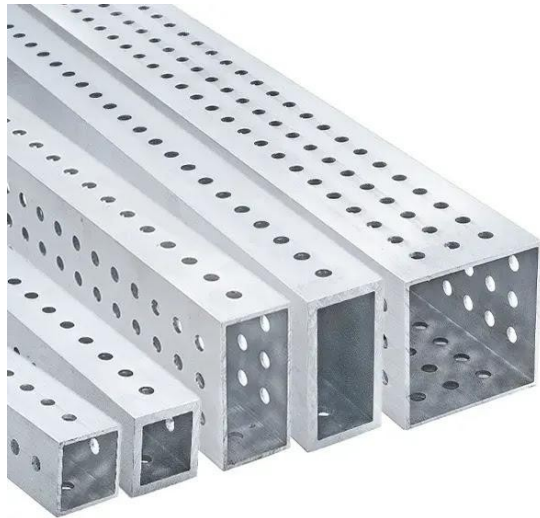
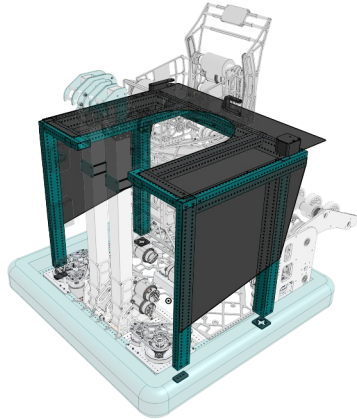
Name: Arnav Krishnan

Report Date: Week of Sep 21

Work Completion Conflicts: *I didn't miss any days but our lead missed Wednesday. We still made progress, but it wasn't the cleanest transition (as we did not know about it ahead of time), and that resulted in some lost time on Wednesday & Friday*

Completed Tasks

Monday - reviewing Onshape, learning [FRCguide.org](https://www.frcguide.org/), starting CAD - 50 minutes

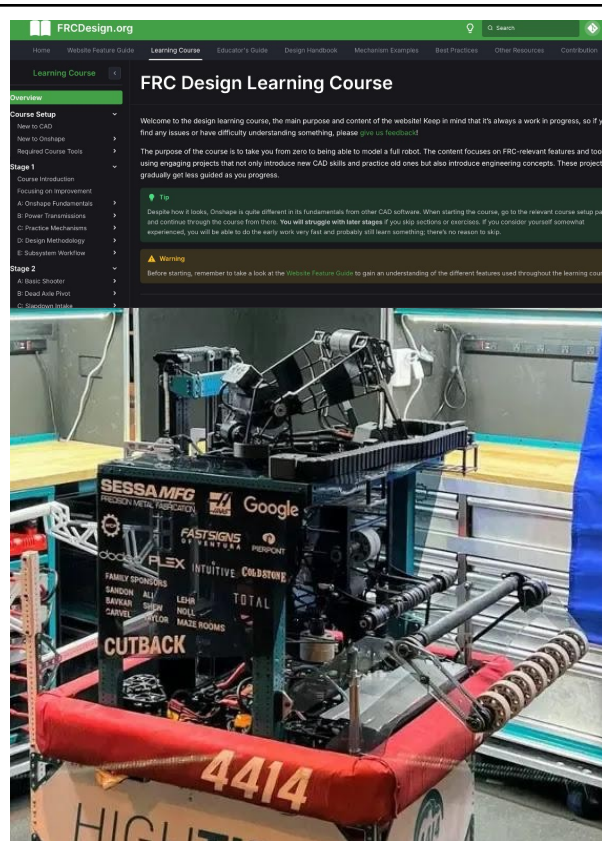


Our team's goal for this unit is to become more capable of designing competition (FRC) robots in CAD, specifically onshape. We were using the [FRCdesign.org](https://www.frcdesign.org/) learning course as the structure because it focuses on tools that are relevant to FRC, and is good for students who are familiar with CAD but less so with CADding a true competition robot (i.e., us).

For Monday, we were mainly getting orientated. We understood the organization of the course, got our Onshape working in Team114, and started the Stage 1A exercise, making a copy of that document.

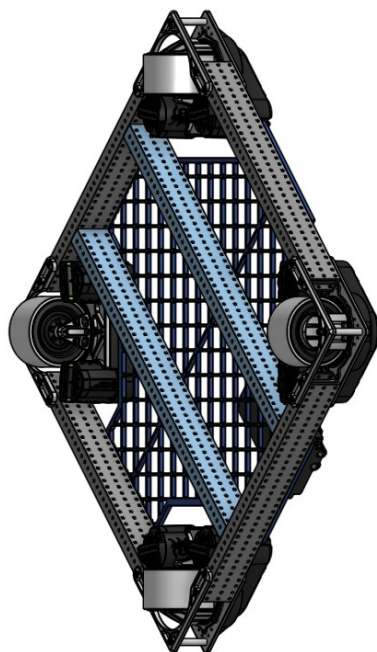
I had some prior Onshape experience, so it was tempting to skip directly ahead to the modeling. However, the course explicitly says that students who do that, skipping ahead, may often struggle. Also, as I began to use Onshape, I realized that my skills were far more 'rusty' than I imagined.

In terms of individual contribution, it is not so pertinent here, as all our work was individual. That being said, my contribution was making a copy of the exercise document, studying the reference material, downloading the necessary tools (like Tube Converter and



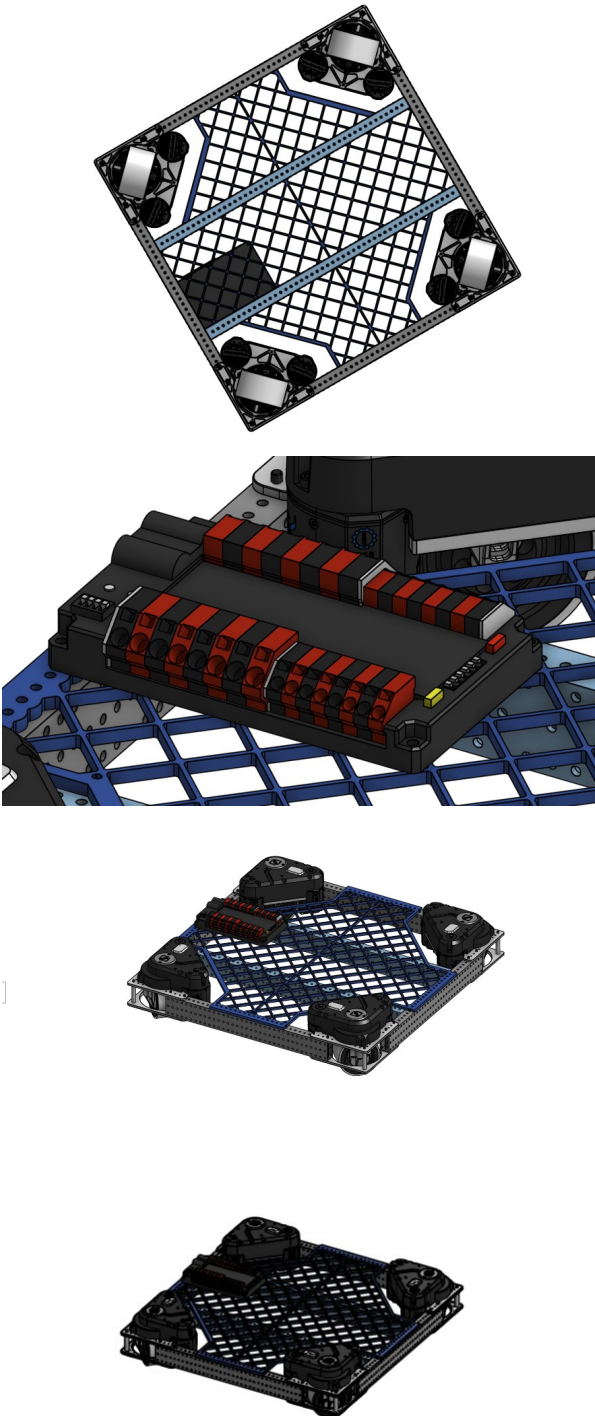
Part Lightning) and beginning the CAD process.

Wednesday & Friday - continue working on CAD (completed) - 3 hours



The Stage 1A course materials include an example document of a swerve drivetrain. Our task was to CAD it, from scratch, in our own document, working on it step by step, and matching each feature to the example. Replicating it is better than just skimming over it because it forces me to make the CAD decisions myself (for example, where to dimension off of, or how to get the right spacing when doing a linear pattern), which will help me in the future when I am doing CAD.

I will briefly go over the steps, but the challenges / lessons are likely more important. I started with the tubes, using a tube converter to add the holes; I then imported 4 motors and used Fastened Mate to join them. Next, I worked on the belly pan, which involved first creating a rectangle. I then used the measure tool to create an aligned square cutout from one

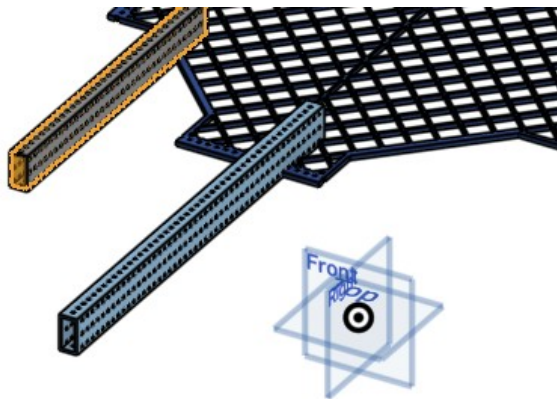
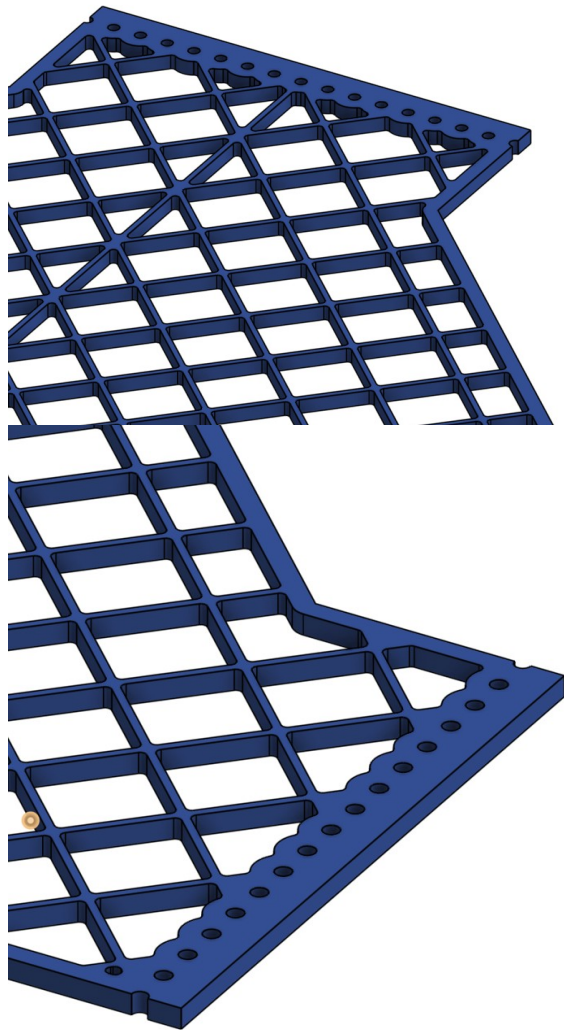


corner, and used the mirror tool twice to take out 4 corners. Finally, I added ribs in the middle. I did this by starting with a single diagonal line, which I enforced to be 0deg with a side of the base which I already knew was 45deg from the the side. I then used a linear pattern to replicate them 30 times with an inch of space, using a mirror over the center to reflect it over the other side. Finally, I used circles to add holes in specific places which I had previously measured, and fastened the power distribution hub to it.

The main challenge I faced was a general one, which was that my skills in CAD had clearly deteriorated. It was definitely a bit frustrating, spending a lot of time “debugging” the CAD. For example, you may notice that in the figure, there is a single vertical rib line going down the center of the belly pan. It seems kind of odd, compared to all the diagonal rib lines. Just minutes before writing this report, I realized my mistake is I forgot to hit “Q” when creating the line to mirror the riblines over, which resulted in the mirror line becoming a rib line as well!

The good news is for most “critical” errors- like the dimensions not adding up- onShape clearly tells you and prompts you to fix it. It’s the more minor, subtle issues like an incorrect rib line that really take a while to fix (and are hence quite frustrating/confusing).

In terms of next steps, next week I am moving onto my last rotation. However, to speak to CAD a bit: if I were to come back, I wish I had more time. It



would have allowed me to really understand the CAD better, maybe even play around more with configurations (for example, what if I adjust the spacing and the # of the linear patterns at the same time?). I have learned a key lesson which is that CAD models should be built such that they are easy to change and replicate, so I wish I could have spent more time making sure my CAD model met those specifications.

